

Penalty Thresholds for the Sensor Placement QUBO: Equality and Inequality Sensor Budgets

Abstract

We characterize the penalty parameter required for exactness of the QUBO reformulation of the water distribution network (WDN) sensor placement problem, for an equality budget $\sum_i x_i = s$ and for an inequality budget $\sum_i x_i \leq s$ encoded with binary slack variables. For arbitrary vertex costs c_i and nonnegative pipe weights w_{ij} , the penalty $\rho > \bar{\rho}_= := \max_i \max\{c_i, W_i - c_i\}$ is strictly exact for the equality budget and $\rho > \bar{\rho}_\leq := \max\{0, \max_i(W_i - c_i)\}$ for the inequality budget, where W_i is the total weight of the pipes at i . Unlike the balanced minimum-cut threshold, these values do not grow with n : a single sensor move changes the objective only through its own neighborhood. Both thresholds are sharp for every instance, because the exact critical penalty attains them at the extreme budgets $s \in \{0, n\}$. For the costs and edge-betweenness weights built by the repository’s **centrality** from nonnegative demands, every network with $n \geq 3$ satisfies $\bar{\rho}_= < 2$, so $\rho = 2$ is exact for every network and every budget, while the rule $\rho = \sum_i |c_i| + 1$ used in the notebook grows from 3.8 to 110 across the networks in the repository. Finally, the exact per-instance critical penalty is an explicit function of the coverage curve, and it vanishes at the budget the notebook selects: there the penalty enforces nothing and only rescales the energy landscape.

1 Setup

Let $G = (V, E)$ be a graph on $n := |V|$ vertices, with a cost $c_i \in \mathbb{R}$ for every vertex and a weight $w_{ij} \geq 0$ for every undirected edge $\{i, j\} \in E$. Write $N(i)$ for the neighbors of i ,

$$W_i := \sum_{j \in N(i)} w_{ij}, \quad a_i := W_i - c_i. \quad (1)$$

Objective. For $\mathbf{x} \in \{0, 1\}^n$, where $x_i = 1$ places a sensor at i , set

$$f(\mathbf{x}) := \sum_{i \in V} c_i x_i + \sum_{\{i, j\} \in E} w_{ij} (1 - x_i)(1 - x_j). \quad (2)$$

A pipe contributes its weight exactly when neither endpoint carries a sensor, so f trades the cost of the sensors against the weight of the pipes left unmonitored. This is the objective of **MIQP**, **QUBO** and **build_Q_matrix** in **src/formulations.py**. Write $S(\mathbf{x}) := \{i : x_i = 1\}$ and $k := |S(\mathbf{x})|$. For $i \in S$, let $R_i^-(\mathbf{x})$ flip x_i from 1 to 0; for $i \notin S$, let $R_i^+(\mathbf{x})$ flip x_i from 0 to 1.

Coverage curves. For $0 \leq k \leq n$ define the equality and inequality optima

$$m_k := \min\{f(\mathbf{x}) : \sum_i x_i = k\}, \quad M_k := \min\{f(\mathbf{x}) : \sum_i x_i \leq k\} = \min_{j \leq k} m_j. \quad (3)$$

The sequence (M_k) is the curve computed by **coverage** in **src/experiments.py**, whose **MIQP** uses $\sum_i x_i \leq s$.

Equality budget. For a budget $0 \leq s \leq n$, the constrained problem has optimal value m_s and the penalized QUBO is

$$F_\rho(\mathbf{x}) := f(\mathbf{x}) + \rho \left(\sum_{i \in V} x_i - s \right)^2. \quad (4)$$

This is $\mathbf{x}^\top Q \mathbf{x} + c_Q$ with (Q, c_Q) returned by `build_Q_matrix`.

Inequality budget. For $s \geq 0$, the constrained problem $\min\{f(\mathbf{x}) : \sum_i x_i \leq s\}$ has optimal value M_s . Introduce K slack bits $\mathbf{y} \in \{0, 1\}^K$ and positive integer coefficients $\beta_1, \dots, \beta_K$ such that

$$B(\mathbf{y}) := \sum_{\ell=1}^K \beta_\ell y_\ell, \quad \{B(\mathbf{y}) : \mathbf{y} \in \{0, 1\}^K\} \supseteq \{0, 1, \dots, s\}. \quad (5)$$

The bounded binary encoding $K = \lfloor \log_2 s \rfloor + 1$, $\beta_\ell = 2^{\ell-1}$ for $\ell < K$ and $\beta_K = s - 2^{K-1} + 1$ has range exactly $\{0, \dots, s\}$; for $s = 0$, $K = 0$. The penalized QUBO is

$$G_\rho(\mathbf{x}, \mathbf{y}) := f(\mathbf{x}) + \rho \left(\sum_{i \in V} x_i + B(\mathbf{y}) - s \right)^2. \quad (6)$$

Write $d(\mathbf{x}, \mathbf{y}) := \sum_i x_i + B(\mathbf{y}) - s$ for the residual of the slack equation.

Goal. We seek penalties guaranteeing *strict exactness*: $F_\rho(\mathbf{x}) > m_s$ for every $\mathbf{x}$ with $\sum_i x_i \neq s$, and $G_\rho(\mathbf{x}, \mathbf{y}) > M_s$ for every $(\mathbf{x}, \mathbf{y})$ with $d(\mathbf{x}, \mathbf{y}) \neq 0$. The per-instance critical penalties are

$$\begin{aligned} \rho_{\leq}^*(s) &:= \inf\{\rho > 0 : \text{every minimizer of } F_\rho \text{ is feasible}\}, \\ \rho_{\leq}^*(s) &:= \inf\{\rho > 0 : \text{every minimizer of } G_\rho \text{ has } d = 0\}. \end{aligned} \quad (7)$$

Remark 1 (Spin units). With $x_i = (z_i + 1)/2$ one has $\sum_i z_i = 2 \sum_i x_i - n$, so $\rho(\sum_i x_i - s)^2 = \rho_z(\sum_i z_i - 2s + n)^2$ with $\rho_z = \rho/4$. All thresholds below are stated for the binary parameter ρ , as in the code; the spin values are obtained by dividing by 4, as in the balanced minimum-cut analysis of [3].

2 One-flip changes

Lemma 2 (One-flip changes). For every $\mathbf{x} \in \{0, 1\}^n$ and $i \in V$,

$$f(R_i^+(\mathbf{x})) - f(\mathbf{x}) = c_i - \sum_{j \in N(i): x_j=0} w_{ij} \in [c_i - W_i, c_i] \quad \text{if } x_i = 0, \quad (8)$$

$$f(R_i^-(\mathbf{x})) - f(\mathbf{x}) = -c_i + \sum_{j \in N(i): x_j=0} w_{ij} \in [-c_i, a_i] \quad \text{if } x_i = 1. \quad (9)$$

Proof. Only the terms of (2) that contain x_i change: $c_i x_i$ and $w_{ij}(1 - x_i)(1 - x_j)$ for $j \in N(i)$. The latter equals w_{ij} when $x_i = x_j = 0$ and 0 otherwise. Placing a sensor at i therefore adds c_i and removes the weight of every pipe from i to an empty neighbor, while pipes to occupied neighbors stay monitored; removing it reverses both effects. The intervals follow from $0 \leq \sum_{j \in N(i): x_j=0} w_{ij} \leq W_i$. $\square$

The change in Lemma 2 depends only on the neighborhood of i . This is the structural difference from the balanced minimum-cut objective of [3], where removing a vertex from a set of size k changes the cut by up to $2(k - 1)$, which grows with n and forces the threshold $\rho > n$.

Lemma 3 (Incident edge betweenness). *Let w_{ij} be the normalized edge betweenness,*

$$w_{ij} = \frac{2}{n(n-1)} \sum_{\{u,v\} \subseteq V} \frac{\sigma_{uv}(ij)}{\sigma_{uv}},$$

where σ_{uv} counts the shortest u - v paths, $\sigma_{uv}(ij)$ those using $\{i, j\}$, and pairs in different components contribute 0. This is `networkx.edge_betweenness_centrality` as called by `centrality`. Then $W_i \leq 2(n-1)/n < 2$ for every i , with equality at the center of a star.

Proof. Summing over the edges at i , $W_i = \frac{2}{n(n-1)} \sum_{\{u,v\}} \sigma_{uv}^{-1} \sum_P |P \cap \delta(i)|$, where P ranges over the shortest u - v paths and $\delta(i)$ is the set of edges at i . A shortest path is simple, so it contains at most one edge at i when i is an endpoint and at most two otherwise. There are $n-1$ pairs containing i and $\binom{n-1}{2}$ pairs not containing it, so

$$W_i \leq \frac{2}{n(n-1)} \left[(n-1) + 2 \binom{n-1}{2} \right] = \frac{2}{n(n-1)} (n-1)(n-1) = \frac{2(n-1)}{n}.$$

At the center of a star every pair containing the center uses exactly one edge at it and every pair of leaves passes through it using exactly two, so both bounds are attained. $\square$

3 Equality budget

Theorem 4 (Strict exactness for $\rho > \bar{\rho}_=$). *Let $0 \leq s \leq n$ and*

$$\bar{\rho}_= := \max_{i \in V} \max\{c_i, W_i - c_i\}. \quad (10)$$

If $\rho > \bar{\rho}_=$, then $\arg \min_{\mathbf{x} \in \{0,1\}^n} F_\rho = \arg \min\{f(\mathbf{x}) : \sum_i x_i = s\}$; in particular every minimizer of F_ρ is feasible and $F_\rho(\mathbf{x}) > m_s$ for every infeasible $\mathbf{x}$. Note that $\bar{\rho}_= \geq \max_i W_i/2 \geq 0$.

Proof. On feasible points the penalty vanishes, so $\min_{\mathbf{x}} F_\rho \leq m_s$. Suppose $\tilde{\mathbf{x}}$ minimizes F_ρ and is infeasible, with $\tilde{S} := S(\tilde{\mathbf{x}})$ and $k = |\tilde{S}| \neq s$.

If $k = s + r$, $r \geq 1$, pick $i \in \tilde{S}$ and set $\mathbf{x}' = R_i^-(\tilde{\mathbf{x}})$. By (9), $f(\mathbf{x}') - f(\tilde{\mathbf{x}}) \leq a_i \leq \bar{\rho}_=$. The penalty changes by $\rho[(r-1)^2 - r^2] = -\rho(2r-1)$, hence

$$F_\rho(\mathbf{x}') - F_\rho(\tilde{\mathbf{x}}) \leq a_i - \rho(2r-1) < \bar{\rho}_= - \bar{\rho}_=(2r-1) = -2\bar{\rho}_=(r-1) \leq 0, \quad (11)$$

where the strict inequality uses $\rho > \bar{\rho}_=$ and $2r-1 \geq 1$. This contradicts the optimality of $\tilde{\mathbf{x}}$. If $k = s - r$, $r \geq 1$, then $k < n$, so some $i \notin \tilde{S}$ exists; set $\mathbf{x}' = R_i^+(\tilde{\mathbf{x}})$. By (8), $f(\mathbf{x}') - f(\tilde{\mathbf{x}}) \leq c_i \leq \bar{\rho}_=$, the penalty drop is again $\rho(2r-1)$, and (11) holds with c_i in place of a_i .

Thus every minimizer is feasible; on the feasible set $F_\rho = f$, so the minimizers of F_ρ are exactly those of the constrained problem, and any infeasible $\mathbf{x}$ satisfies $F_\rho(\mathbf{x}) > \min F_\rho = m_s$. $\square$

Theorem 5 (Boundary case $\rho = \bar{\rho}_=$). *Let $\rho = \bar{\rho}_=$ and $\tilde{\mathbf{x}} \in \arg \min F_{\bar{\rho}_=}$. A finite sequence of one-bit repairs (R_i^- while $\sum_i x_i > s$, R_i^+ while $\sum_i x_i < s$) produces a feasible $\mathbf{x}^0$ with $F_{\bar{\rho}_=}(x^0) \leq F_{\bar{\rho}_=}(x)$ and $f(x^0) = m_s$. Hence $\min F_{\bar{\rho}_=} = m_s$, although infeasible optimal ties may occur.*

Proof. For any $\mathbf{x}$ with $\sum_i x_i = s + r$, $r \geq 1$, and $i \in S(\mathbf{x})$, (11) with $\rho = \bar{\rho}_=$ gives $F_{\bar{\rho}_=}(R_i^-(\mathbf{x})) - F_{\bar{\rho}_=}(x) \leq -2\bar{\rho}_=(r-1) \leq 0$; the case $\sum_i x_i = s - r$ is analogous. Each step is nonworsening and reduces $|\sum_i x_i - s|$ by one, so a feasible $\mathbf{x}^0$ with $F_{\bar{\rho}_=}(x^0) \leq F_{\bar{\rho}_=}(x) \leq m_s$ is reached, and feasibility forces $f(x^0) = F_{\bar{\rho}_=}(x^0) \geq m_s$. $\square$

The proof of Theorem 4 only needs *one* repairable vertex, which yields a budget-dependent refinement. Write $a_{[1]} \geq \dots \geq a_{[n]}$ and $c_{[1]} \geq \dots \geq c_{[n]}$ for the sorted values, and set $a_{[n+1]} = c_{[n+1]} := -\infty$.

Corollary 6 (Budget-dependent threshold). *Let*

$$\bar{\rho}_=(s) := \max\{0, a_{[s+1]}, c_{[n-s+1]}\} \leq \bar{\rho}_=. \quad (12)$$

If $\rho > \bar{\rho}_=(s)$, the conclusions of Theorem 4 hold.

Proof. If $k = s + r$, the set $\tilde{S}$ has at least $s + 1$ elements, so it contains some i with $a_i \leq a_{[s+1]}$; repairing that i gives $F_\rho(\mathbf{x}') - F_\rho(\tilde{\mathbf{x}}) \leq a_{[s+1]} - \rho(2r - 1) \leq a_{[s+1]} - \rho < 0$. If $k = s - r$, the complement $V \setminus \tilde{S}$ has at least $n - s + 1$ elements, so it contains some i with $c_i \leq c_{[n-s+1]}$, and the same estimate holds. The inequality $\bar{\rho}_=(s) \leq \bar{\rho}_=$ is immediate, since $a_{[s+1]} \leq \max_i a_i$, $c_{[n-s+1]} \leq \max_i c_i$ and $\bar{\rho}_= \geq 0$. $\square$

Proposition 7 (Sharpness of the equality threshold).

- (a) Every instance attains $\bar{\rho}_=$. For every instance, $\rho_-^*(0) = \max\{0, \max_i a_i\}$ and $\rho_-^*(n) = \max\{0, \max_i c_i\}$. Consequently

$$\bar{\rho}_= = \max_{0 \leq s \leq n} \rho_-^*(s),$$

so no budget-independent penalty below $\bar{\rho}_=$ is exact for every budget of a given network, and Corollary 6 is attained at $s \in \{0, n\}$.

- (b) Pipe term, every budget. For $s \geq 0$ and $w > 0$, let G consist of $s + 1$ disjoint pipes of weight w with $c \equiv 0$. Then $\bar{\rho}_= = w$, and for every $\rho < w$ every minimizer of F_ρ is infeasible; at $\rho = w$ infeasible optimal ties occur.
- (c) Cost term, every budget. For $s \geq 1$ and $c > 0$, let G consist of $n = s + 1$ isolated vertices of cost c . Then $\bar{\rho}_=(s) = \bar{\rho}_= = c$, and for every $\rho < c$ every minimizer of F_ρ is infeasible.

Proof. (a) For $s = 0$ the only feasible point is $\mathbf{x} = \mathbf{0}$, with $m_0 = \sum_{\{i,j\} \in E} w_{ij}$. Placing one sensor at i gives, by (8) with every neighbor empty, $F_\rho = m_0 - a_i + \rho$, which beats m_0 whenever $\rho < a_i$; hence $\rho_-^*(0) \geq \max\{0, \max_i a_i\}$, and Corollary 6 at $s = 0$ gives the matching upper bound. For $s = n$ the only feasible point is $\mathbf{1}$, with $m_n = \sum_i c_i$; removing the sensor at i leaves every pipe monitored, so $F_\rho = m_n - c_i + \rho$, and the same two arguments give $\rho_-^*(n) = \max\{0, \max_i c_i\}$. Finally $\bar{\rho}_= = \max\{\rho_-^*(0), \rho_-^*(n)\} \leq \max_s \rho_-^*(s) \leq \bar{\rho}_=$, the last inequality by Theorem 4.

(b) Here $n = 2s + 2$, $W_i = w$ and $c_i = 0$, so $\bar{\rho}_= = w$. A feasible point places s sensors and monitors at most s pipes, so $m_s = w$, attained with one sensor on each of s pipes. The point $\mathbf{x}^\bullet$ with one sensor on every pipe has $f(\mathbf{x}^\bullet) = 0$ and $F_\rho(\mathbf{x}^\bullet) = \rho < w = m_s$, so $\mathbf{x}^\bullet$ beats every feasible point and all minimizers are infeasible. At $\rho = w$, $F_w(\mathbf{x}^\bullet) = m_s$ is an infeasible tie.

(c) There are no pipes, so $a_i = -c$, $c_{[n-s+1]} = c_{[2]} = c$ and $\bar{\rho}_=(s) = \bar{\rho}_= = c$. Feasible points cost $m_s = sc$, while $s - 1$ sensors give $F_\rho = (s - 1)c + \rho < sc$ for $\rho < c$. $\square$

4 Inequality budget

Theorem 8 (Strict exactness for $\rho > \bar{\rho}_\leq$). *Let $s \geq 0$, let B satisfy (5), and let*

$$\bar{\rho}_\leq := \max\left\{0, \max_{i \in V} (W_i - c_i)\right\}. \quad (13)$$

If $\rho > \bar{\rho}_\leq$, then every minimizer $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ of G_ρ satisfies $d(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) = 0$, hence $\sum_i \tilde{x}_i \leq s$; moreover $\min G_\rho = M_s$, and the $\mathbf{x}$ -parts of the minimizers of G_ρ are exactly the minimizers of $\{f(\mathbf{x}) : \sum_i x_i \leq s\}$.

Proof. For every $\mathbf{x}$ with $\sum_i x_i \leq s$, (5) provides $\mathbf{y}_\mathbf{x}$ with $B(\mathbf{y}_\mathbf{x}) = s - \sum_i x_i$, so $G_\rho(\mathbf{x}, \mathbf{y}_\mathbf{x}) = f(\mathbf{x})$ and $\min G_\rho \leq M_s$. Suppose $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ minimizes G_ρ with $\tilde{d} := d(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \neq 0$.

If $\sum_i \tilde{x}_i \leq s$, then $G_\rho(\tilde{\mathbf{x}}, \mathbf{y}_\mathbf{x}) = f(\tilde{\mathbf{x}}) < f(\tilde{\mathbf{x}}) + \rho \tilde{d}^2 = G_\rho(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ because $\rho > 0$, a contradiction.

If $\sum_i \tilde{x}_i = s + r$ with $r \geq 1$, then $\tilde{d} \geq r \geq 1$ because $B \geq 0$. Pick $i \in S(\tilde{\mathbf{x}})$, set $\mathbf{x}' = R_i^-(\tilde{\mathbf{x}})$ and keep $\tilde{\mathbf{y}}$. By (9), $f(\mathbf{x}') - f(\tilde{\mathbf{x}}) \leq a_i \leq \bar{\rho}_\leq$, and the penalty changes by $\rho[(\tilde{d} - 1)^2 - \tilde{d}^2] = -\rho(2\tilde{d} - 1)$, so

$$G_\rho(\mathbf{x}', \tilde{\mathbf{y}}) - G_\rho(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \leq a_i - \rho(2\tilde{d} - 1) < \bar{\rho}_\leq - \bar{\rho}_\leq(2\tilde{d} - 1) \leq 0, \quad (14)$$

again a contradiction. Hence every minimizer has $d = 0$ and thus $\sum_i \tilde{x}_i \leq s$ and $G_\rho(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) = f(\tilde{\mathbf{x}}) \geq M_s$; combined with $\min G_\rho \leq M_s$ this gives $\min G_\rho = M_s$ and the optimality of $\tilde{\mathbf{x}}$. Conversely every optimal $\mathbf{x}$ extends to the minimizer $(\mathbf{x}, \mathbf{y}_\mathbf{x})$. $\square$

Theorem 9 (Boundary case $\rho = \bar{\rho}_\leq$). *If $\bar{\rho}_\leq > 0$, then $\min G_{\bar{\rho}_\leq} = M_s$, although infeasible optimal ties may occur.*

Proof. Let $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ be a minimizer. While $\sum_i x_i > s$ we have $d \geq 1$, and (14) with $\rho = \bar{\rho}_\leq$ shows that removing any sensor does not increase $G_{\bar{\rho}_\leq}$. After finitely many removals $\sum_i x_i \leq s$, and replacing $\mathbf{y}$ by $\mathbf{y}_\mathbf{x}$ gives a point with value $f(\mathbf{x}) \geq M_s$ and no larger than $G_{\bar{\rho}_\leq}(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \leq M_s$. $\square$

Corollary 10 (Budget-dependent threshold). *If $\rho > \bar{\rho}_\leq(s) := \max\{0, a_{[s+1]}\} \leq \bar{\rho}_\leq$, the conclusions of Theorem 8 hold.*

Proof. In the case $\sum_i \tilde{x}_i = s + r$, the set $S(\tilde{\mathbf{x}})$ has at least $s + 1$ elements and contains some i with $a_i \leq a_{[s+1]}$; repair that vertex in (14). $\square$

Proposition 11 (Sharpness of the inequality threshold). *(a) For every instance, $\rho_\leq^*(0) = \bar{\rho}_\leq$, so $\bar{\rho}_\leq = \max_{s \geq 0} \rho_\leq^*(s)$. (b) For the $s + 1$ disjoint pipes of Proposition 7(b), $\bar{\rho}_\leq = w$, $M_s = w$, and for every $\rho < w$ every minimizer of G_ρ places more than s sensors.*

Proof. (a) For $s = 0$ the encoding is empty ($K = 0$), so $G_\rho = F_\rho$ and $\rho_\leq^*(0) = \rho_\leq^*(0) = \max\{0, \max_i a_i\} = \bar{\rho}_\leq$ by Proposition 7(a); Theorem 8 gives $\rho_\leq^*(s) \leq \bar{\rho}_\leq$ for every s . (b) A point with at most s sensors monitors at most s pipes, so $M_s = w$. The point $\mathbf{x}^\bullet$ with one sensor per pipe has $\min_{\mathbf{y}} G_\rho(\mathbf{x}^\bullet, \mathbf{y}) = \rho$, attained at $\mathbf{y} = \mathbf{0}$, which is below M_s when $\rho < w$; hence no point with at most s sensors is a minimizer. $\square$

Remark 12 (Why the cost term disappears). *Under the inequality budget, a deficit $\sum_i x_i < s$ is absorbed by the slack at no cost, so only the removal direction (9) must be repaired, and c_i no longer enters the threshold. Equivalently, for every $\mathbf{x}$ and $\rho > 0$, $\min_{\mathbf{y}} G_\rho(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) + \rho \max\{0, \sum_i x_i - s\}^2$: the slack turns the two-sided penalty into a one-sided one. The price is $K = \lfloor \log_2 s \rfloor + 1$ extra variables, coupled to every x_i with strength $\rho\beta_\ell$, so the QUBO remains a complete graph on $n + K$ variables.*

5 Water distribution networks

The repository builds c and w with `centrality` in `src/network.py`: $c_i = q_i / \max_j q_j + \deg(i) / (n - 1)^2$, where $q_i \geq 0$ is the water demand at i and $\deg(i) / (n - 1)$ is `networkx`'s degree centrality divided by $n - 1$ once more, and w_{ij} is the normalized edge betweenness of Lemma 3.

Corollary 13 (A size-independent exact penalty). *Assume $q_i \geq 0$ for all i and $\max_j q_j > 0$. Then $0 \leq c_i \leq 1 + \frac{1}{n-1}$ and $W_i \leq \frac{2(n-1)}{n}$, so*

$$\bar{\rho}_= \leq \max\left\{1 + \frac{1}{n-1}, \frac{2(n-1)}{n}\right\}, \quad \bar{\rho}_\leq \leq \frac{2(n-1)}{n}.$$

In particular, for $n \geq 3$ the penalty $\rho = 2$ is strictly exact for every network, every budget s , and both the equality and the inequality budget. The notebook's rule $\rho_{\text{nb}} := \sum_i |c_i| + 1$ satisfies $\rho_{\text{nb}} \geq 2$ and is therefore also exact for $n \geq 3$, but $\rho_{\text{nb}} \geq 1 + \sum_i q_i / \max_j q_j$ grows linearly in n for comparable demands.

Proof. The demand term lies in $[0, 1]$ and the degree term in $[0, \frac{1}{n-1}]$, since $\deg(i) \leq n-1$. Lemma 3 bounds W_i , and $W_i - c_i \leq W_i$. For $n \geq 3$, $1 + \frac{1}{n-1} \leq \frac{3}{2} < 2$ and $\frac{2(n-1)}{n} < 2$, so Theorems 4 and 8 apply at $\rho = 2$. If i^* has the largest demand then $c_{i^*} \geq 1$, so $\rho_{\text{nb}} \geq c_{i^*} + 1 \geq 2$. $\square$

Remark 14 (Degree term). *If the second division by $n-1$ is removed, so that the degree term is networkx 's degree centrality in $[0, 1]$ (SUSPECT-1 in `docs/code_review.md`), the same argument gives $c_i \leq 2$ and $\bar{\rho}_= \leq 2$, so every $\rho > 2$ is exact.*

Table 1 evaluates the thresholds on every network in `Data/WDN_Data`. In every network $\bar{\rho}_= = \max_i c_i \approx 1$, set by the vertex with the largest demand, regardless of size, while ρ_{nb} is 4 to 110 times larger.

Network	n	$ E $	$\max_i c_i$	$\max_i W_i$	$\frac{2(n-1)}{n}$	ρ_{nb}	$\bar{\rho}_=$	$\bar{\rho}_\leq$
Test	18	18	1.0069	0.2353	1.8889	3.821	1.0069	0.1636
Apulia	24	34	1.0057	0.5709	1.9167	4.473	1.0057	0.4493
Fossolo	37	58	1.0023	0.4205	1.9459	5.581	1.0023	0.3386
ZJ	114	162	1.0002	1.0506	1.9825	23.659	1.0002	0.9130
Modena	272	317	1.0000	0.6186	1.9926	43.980	1.0000	0.4311
Kentucky	856	982	1.0000	0.5498	1.9977	109.987	1.0000	0.4944

Table 1: Penalty thresholds for the networks in `Data/WDN_Data`, with costs and weights from `centrality`. $\rho_{\text{nb}} = \sum_i |c_i| + 1$ is the value used in the notebook and stored in `logs/`. Values are rounded to four decimals; $\max_i c_i$ exceeds 1 only through the degree term, which is at most $1/(n-1)$.

6 Per-instance critical penalty

Because the configuration space is finite, the critical penalties are explicit functions of the coverage curves (3), in the same way as the identity for ρ^* in the balanced minimum-cut analysis of [3].

Proposition 15 (Critical penalty from the coverage curve). *For every instance,*

$$\rho_-^*(s) = \max\left\{0, \max_{k \neq s} \frac{m_s - m_k}{(k - s)^2}\right\}, \quad (15)$$

$$\rho_\leq^*(s) = \max\left\{0, \max_{k > s} \frac{M_s - M_k}{(k - s)^2}\right\}. \quad (16)$$

Proof. For (15): every minimizer of F_ρ is feasible exactly when $f(\mathbf{x}) + \rho(\sum_i x_i - s)^2 > m_s$ for every infeasible $\mathbf{x}$. For a fixed count $k \neq s$ the most demanding $\mathbf{x}$ is one with $f(\mathbf{x}) = m_k$, so the condition reads $\rho > (m_s - m_k)/(k - s)^2$ for all $k \neq s$; at equality an infeasible tie exists.

For (16): if $\sum_i x_i \leq s$ then $\min_{\mathbf{y}} G_\rho(\mathbf{x}, \mathbf{y}) = f(\mathbf{x})$, attained only where $d = 0$ when $\rho > 0$. If $\sum_i x_i = k > s$ then $d \geq k - s$ with equality at $\mathbf{y} = \mathbf{0}$, so $\min_{\mathbf{y}} G_\rho(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) + \rho(k - s)^2$. Hence every minimizer has $d = 0$ if and only if $\rho > 0$ and $\rho > (M_s - m_k)/(k - s)^2$ for every $k > s$. Replacing m_k by $M_k \leq m_k$ cannot lower any term; conversely, if $M_k = m_j$ with $s < j \leq k$ then $(M_s - M_k)/(k - s)^2 \leq (M_s - m_j)/(j - s)^2$, and if $M_k = M_s$ the term is 0. The two maxima therefore coincide. $\square$

The notebook chooses the budget as the first sensor count at which the coverage curve reaches its minimum (cell 14). At that budget no penalty is needed.

Proposition 16 (Budget at the minimum of the coverage curve). *Let s^* be the smallest minimizer of $k \mapsto M_k$. Then $m_{s^*} = M_{s^*} = \min_k m_k$, and $\rho_-^*(s^*) = \rho_{\leq}^*(s^*) = 0$: every $\rho > 0$ is strictly exact at s^* , for both budgets.*

Proof. The sequence (M_k) is nonincreasing with $M_n = \min_k m_k$. If $m_{s^*} > M_{s^*}$ then $s^* \geq 1$ and $M_{s^*-1} = M_{s^*}$, contradicting the minimality of s^* . Hence $m_{s^*} = \min_k m_k$, every term $m_{s^*} - m_k$ in (15) is nonpositive, and $M_{s^*} - M_k = 0$ for all $k > s^*$ in (16). $\square$

At s^* the unconstrained minimum of f already places s^* sensors, and the penalty only breaks ties between budgets. Its magnitude then plays no role in exactness and only rescales the landscape seen by simulated annealing, tabu search and the annealer.

Exact values, numerically. For the two networks small enough to enumerate, Table 2 reports the exact critical penalties over all budgets, computed from (15) and (16) with the coverage curves obtained by evaluating all 2^n placements. For Apulia, $s^* = 11$ is the budget stored in `logs/Apulia`, and $m_{11} = 1.300182$ equals, to ten digits, the best tabu search energy in `Apulia_TabuSearch.json`.

Network	n	s^*	m_{s^*}	$\rho_-^*(s^*)$	$\max_{0 < s < n} \rho_-^*(s)$	$\text{med}_s \rho_-^*(s)$	$\rho_-^*(n) = \bar{\rho}_=$	$\rho_{\leq}^*(0) = \bar{\rho}_{\leq}$
Test	18	6	0.8115	0	0.1418	0.1210	1.0069	0.1636
Apulia	24	11	1.3002	0	0.3546	0.1109	1.0057	0.4493

Table 2: Exact critical penalties by enumeration of all 2^n placements. The maximum over intermediate budgets is attained at $s = 1$ in both networks, and $\rho_{\leq}^*(s) = 0$ for every $s \geq s^*$.

Remark 17 (Computing ρ^* at scale). *Proposition 15 reduces ρ^* to $n + 1$ constrained optima. For $n \lesssim 25$ they follow from exhaustive enumeration, as above. For larger networks they are $n + 1$ MIQP solves with $\sum_i x_i = k$; for the inequality budget the curve (M_k) returned by `coverage` already suffices. Since `coverage` uses $\sum_i x_i \leq s$ (BUG-03 in `docs/code_review.md`), it returns M , not m , so (15) requires the equality version.*

7 Synthesis and open problems

Three penalties describe the same QUBO:

- **Instance threshold.** $\rho > \bar{\rho}_=$ (equality) or $\rho > \bar{\rho}_\leq$ (inequality) is exact for every budget (Theorems 4–9), and no smaller budget-independent penalty is (Propositions 7 and 11). The thresholds depend only on the neighborhood of each vertex, and for the repository’s networks they are below 2 independently of n (Corollary 13).
- **Per-instance critical penalty.** $\rho^*(s)$ is an explicit function of the coverage curve (Proposition 15); it vanishes at the budget the notebook selects (Proposition 16) and stays below 0.36 at every intermediate budget of the enumerated networks.
- **Notebook rule.** $\rho_{\text{nb}} = \sum_i |c_i| + 1$ is exact but grows linearly in n , reaching 110 on Kentucky.

There is no conflict between these values: ρ_{nb} is exact because it exceeds $\bar{\rho}_=$, and $\bar{\rho}_=$ is the worst case of $\rho^*(s)$ over budgets.

Practical guidance. Use $\rho = 2$ for every network with $n \geq 3$, or $\bar{\rho}_=(s)(1 + \varepsilon)$ from Corollary 6 when a smaller value is wanted for a fixed budget. At s^* , any small $\rho > 0$ is exact. The penalty sets the scale of Q : the diagonal entries are $c_i - W_i + \rho(1 - 2s)$, which for Modena ($s = 87$) is about -7608 at $\rho_{\text{nb}} = 43.98$, the value printed in the notebook, against about -346 at $\rho = 2$, a dynamic range 22 times smaller. Whether this improves the samplers is an empirical question that the logs cannot answer, since every run used ρ_{nb} ; it is most pressing for quantum annealing, which found no feasible sample on ZJ.

Open problems. First, the effect of ρ on heuristic and quantum samplers, as opposed to exactness, which is settled here: a smaller penalty flattens the barriers between budgets, which may help or hurt mixing. Second, encodings of the inequality budget without slack bits, such as one-sided penalties that are not quadratic, are not covered by Theorem 8. Third, problems with several budgets, for example per-district sensor limits, add one penalty per constraint; the one-flip argument extends when every vertex appears in a bounded number of constraints, but the interaction between penalties is not analyzed here.

Relation to prior work. The one-bit repair arguments follow the QUBO exactness pattern of Quintero, Bernal, Terlaky, and Zuluaga [2], and the structure of this note follows the balanced minimum-cut analysis of [3]. Slack-variable encodings of inequality constraints in QUBOs are standard; see [1].

Reproducibility. Every statement above is checked numerically by `docs/penalty_parameter/verify_penalty_thresholds.py`. Run it from the repository root. It checks:

- Lemmas 2–3 on random graphs;
- Theorems 4–9, Corollaries 6 and 10, Propositions 7(a), 11(a), 15 and 16, by exhaustive enumeration of the penalized QUBOs, including the slack bits, on 300 random connected graphs with $n \leq 9$ (seed 20261007), half scored with `centrality` and half with generic $c_i \sim U[-1, 2]$, $w_{ij} \sim U[0, 2]$;
- the sharpness families of Propositions 7(b)–(c) and 11(b).

It also computes Tables 1 and 2, which it writes to `penalty_thresholds_results.json`.

References

- [1] A. Lucas, “Ising formulations of many NP problems,” *Frontiers in Physics*, 2 (2014), p. 5.
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